Joseph Mills

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SUMMARY

Computer Science student working as a software engineer and machine learning research assistant. Built a custom Chromium-based browser with two public releases, a deployed LLM translation app, and a published open-source ML benchmark. Former Team Alberta biathlete and youth coach who works well under pressure and enjoys explaining technical work to others.

SKILLS

Languages: Python, Java, Go, TypeScript, JavaScript, SQL, C++, Rust, C, Haskell, R, C#
Machine Learning, AI & Data: PyTorch, ONNX, TensorFlow Lite, LangGraph, Gemini, RAG, Qdrant, pandas, GeoPandas, SciPy, Power BI, Databricks
Web & Backend: React, Next.js, Node.js, Express, FastAPI, REST APIs, Postgres, SQLite, Redis
DevOps & Tools: Docker, CI/CD, GitHub Actions, Linux, Git, PowerShell, Playwright, AWS, Azure, Vercel, Supabase, Kubernetes, Agile/Scrum

EDUCATION

Mount Royal University Sep. 2023 – Dec. 2027 (Expected)
Bachelor of Science in Computer Science | CGPA 3.59 *Calgary, AB*

- Awards: NSERC Undergraduate Student Research Award (USRA), Dean's Honour Roll
- Relevant coursework: Artificial Intelligence, Advanced Software Engineering, Algorithms and Complexity, Data Structures (**Java**), Operating Systems, Databases, Network Infrastructure, Advanced Web Development, Programming Paradigms, Statistics, Linear Algebra for Data Science

EXPERIENCE

Software Engineer and Machine Learning Research Assistant Jan. 2026 – Present
Mount Royal University *Calgary, AB*

- Built SplitBench, an open-source **Python** framework for training AI across many devices without sharing their data, comparing 7 methods on 7 datasets
- Co-authored a paper accepted at IEEE IRI 2026 and took part in regular code reviews with the research supervisor
- Built a **Python** scheduler that splits AI training across devices based on their hardware and network speed, predicting training time within 2%
- Simulated slow machines and networks in 27 **Docker** setups so any experiment reruns with one command, and automated 80+ unattended **PowerShell** runs
- Nearly doubled results on rare classes in medical image data (28% → 52% F1) using **PyTorch**
- Tested 20 model setups on a 7-class medical image dataset, reaching 87.6% accuracy

Data Science Intern Jan. 2026 – Mar. 2026
Miistakis Institute *Calgary, AB*

- Fixed mapping errors across a 260,000-polygon geodatabase with **GeoPandas** so data from different sources lined up
- Interviewed staff to find their slowest manual task, then replaced their multi-day Excel process with a **Python** pipeline that runs in minutes
- Wrote the report used to brief external stakeholders, translating biodiversity statistics (**SciPy**) into plain-language findings

Jump Crew and Signage Team May 2023 – Sep. 2025
Spruce Meadows *Calgary, AB*

- Completed full course changes in 5-minute commercial breaks at live televised events by coordinating the crew
- Planned weekly signage tasks with upper management and trained the incoming summer manager the next season

PROJECTS

Boring Browser | *C++, Rust, Python, Chromium, GitHub Actions* jd-d3v.github.io/boring-web-browser | 2026

- Built a privacy-focused web browser on **Chromium** with 37 **C++** changes to its source; shipped 2 public releases
- Interviewed seniors who had fallen for online scams and surveyed Reddit users to decide which safety features to build
- Added a **Rust** ad-blocking engine to **Chromium**, blocking trackers disguised as part of the site you are visiting
- Automated releases with a **GitHub Actions CI/CD** pipeline, including signed automatic updates
- Tested every build automatically with 30+ **Python** end-to-end test suites covering every major feature
- Added opt-in AI page summaries from 5 **LLM** providers, with encrypted storage of users' API keys

Pipe Dreams | *Python, React, TypeScript, FastAPI, scikit-learn, GeoPandas, ElevenLabs* 2026

- Won **Best Project Built with ElevenLabs** at the IEEE Southern Alberta Industry Hackathon 2026 (26 teams), as a team of 2
- Built the **React/TypeScript** dashboard and **FastAPI** backend for an AI agent that ranks Calgary water mains for inspection, using 37,538 past breaks and 60,740 pipe segments
- Added an **ElevenLabs** voice agent that talks city staff through why each pipe was ranked

Beyonder | *Python, LangGraph, Gemini, FastAPI, Next.js, Postgres, Qdrant, Docker* jd-d3v.github.io/beyonder | 2026

- Built a web-novel translation app using a **LangGraph** AI agent on **Gemini** that reviews and corrects its own translations
- Answered reader questions without spoilers by searching only the chapters they had finished (**Qdrant** semantic search)
- Fixed timeouts on long chapters by translating in resumable chunks, so the app runs on a 512 MB server

Woofer | *JavaScript, TypeScript, Go, Next.js, Postgres* woofer-web-ashen.vercel.app | 2026

- Built a dog adoption app in **Next.js**, live on **Vercel**, where adopters browse dogs and shelters review applicants
- Designed a **Go** REST API on **Postgres** for listings, matches, and messaging, with login required on every endpoint

Classroom Finder | *React, TypeScript, Python, Playwright* jd-d3v.github.io/Class-Finder | 2025

- Built a **React/TypeScript** web app and Telegram bot that read the university timetable to show which classrooms are free
- Wrote the **Python** REST backend serving both apps, matching room requests against class times

Music Browser | *JavaScript, React, Node.js, Express, SQLite* jd-d3v.github.io/COMP4513Assign2 | 2026

- Built and deployed a music catalogue app in **React** with a **Node.js/Express** REST API on **SQLite**, hosted on GitHub Pages and Render
- Added filtering and sorting, charts comparing 6 audio measurements per song, and a feature that finds the 3 most similar tracks

Multi-Threaded Web Server | *C, POSIX Threads, Sockets, Linux* github.com/JSenk347/web-server | 2025

- Built the thread pool and producer-consumer request queue for an HTTP server in **C**, using **pthreads** mutexes and condition variables so 4 workers serve connections in parallel, as part of a team of 3
- Added load shedding that drops connections when the queue is full, plus a live dashboard showing queue size and total requests

ATHLETICS AND ACTIVITIES

Biathlon, Team Alberta 2013 – 2020
Foothills Nordic *Calgary, AB*

- Won gold at the North American Championships and bronze at the Canadian National Championships
- Selected to Team Alberta; attended national team camps and performance testing with national team athletes
- Trained 11 hours per week for 5 years while completing high school full-time

Biathlon Coach 2019 – 2023
Bragg Creek Nordic *Bragg Creek, AB*

- Coached junior athletes for four seasons, running training, supervising range safety, and supporting race days
- Introduced new athletes to the sport and prepared them for their first provincial races

MRUHacks Hackathon Oct. 2025
Mount Royal University *Calgary, AB*

- Built MoJo, a peer-to-peer payment app, in 24 hours as a team of 5 (React Native, React, Flask, MongoDB, JWT, AWS)